

X-Ray Photoelectron Spectroscopy and Auger Electron Spectroscopy

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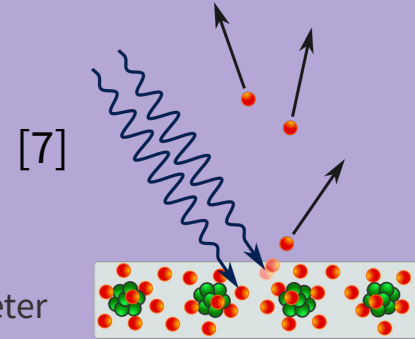


Photoelectric Effect for XPS

- The photoelectric effect - light can remove electrons from an atom
- Conservation of energy equation:

$$h\nu - U - \phi_{\text{sp}} = T$$

- Variables
 - $h\nu$: the energy of the incident x-rays
 - U : binding energy of the photoelectron
 - ϕ_{sp} : the work function, which is fixed for an individual spectrometer
 - T : the kinetic energy of the outgoing photoelectron
- $h\nu$ and ϕ_{sp} are known quantities, so the binding energies can be calculated from measurements of the kinetic energy.
- The histogram of electron binding energies can then be matched to a known spectrum for some material.

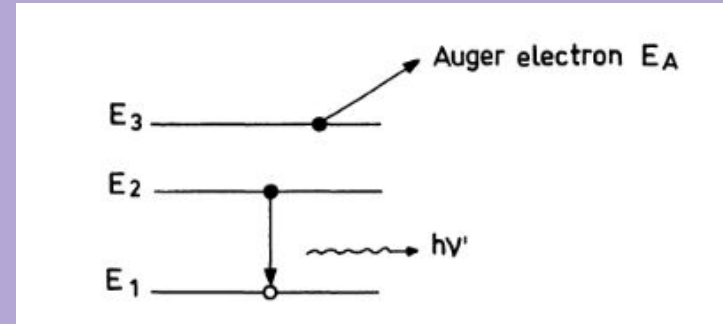


The Auger Effect

- If a core electron is knocked out of an atom, another electron can fill the “hole” left in its place. This will decrease that electron’s potential causing a valence electron(Auger electron) to be emitted to conserve energy.

$$-U_2 - U_3^* = T_A - U_1 \quad [2]$$

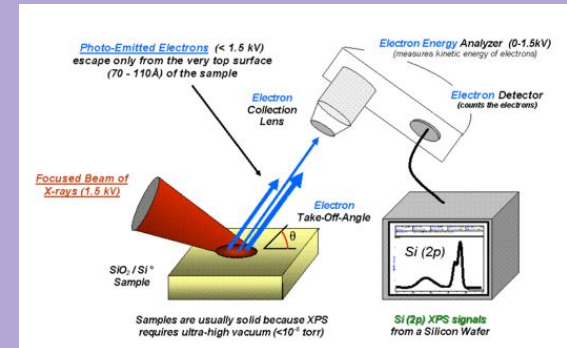
- Variables:
 - U_2 : Binding energy at level 2
 - U_3^* : Binding energy at level 3, assuming a hole in level 1
 - U_1 : Binding energy at level 1
 - T_A : Kinetic energy of the Auger electron



XPS Experimental Process

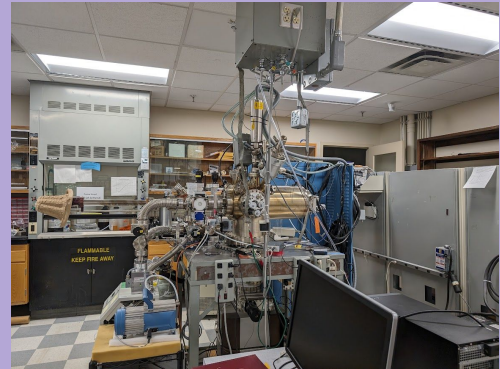
- A beam of X-Ray is focused onto the sample, causing the photoelectric effect.
- The photoelectrons are collected by a device, called a CMA, which determines their kinetic energy.
- The formula from the last slide is used to calculate the binding energy.
- A computer generates a histogram for the frequencies of the binding energies.
- This histogram is then compared with the XPS spectra of known elements.
- The binding energies of an element can be influenced if there is an additional element that it is bound to, appearing as a left or right shift in the spectrum.
 - A shift can be used to determine the the bonds of an element, as these shifts are known.
- Requires extremely low pressure, $\sim 10^{-8}$ Torr

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Uses of XPS

- Used to determine chemical makeup of the surface of a material
- Scan depth: ~5-10nm
- Capable of measuring several aspects of a material, including:
 - Elemental composition
 - Binding energies
 - Layer thickness
- Depth profiling: removes layers and measures them to determine the makeup of each layer
- Provides best data for inorganic materials



Additional Structure of XPS Spectra Examples

Shakeup Satellites

- Loss of the photoelectron causes the increase in the relative charge of the nucleus
- This leads to an energetic reorganization of valence electrons, sometimes sending one to excited states
- This can decrease the energy of the photoelectron, leading to a nearby lower energy peak

Multiplet Splitting

- Occurs in the presence of unpaired valence electrons
- Photoelectron will either be aligned or anti-aligned with valence spin, leading to energy splitting (spin-orbit coupling)

$$\frac{I_+}{I_-} = \frac{S + 1}{S}$$

$$E = (2S + 1)K$$

Additional Structure of XPS Spectra Examples 2

Plasmon Loss

- sometimes photoelectrons trigger plasmons(quantized oscillations), losing discrete amounts of energy in the process
- Peaks intensity decreases as the plasmon energies increase

$$E_p = n\hbar\omega_b$$

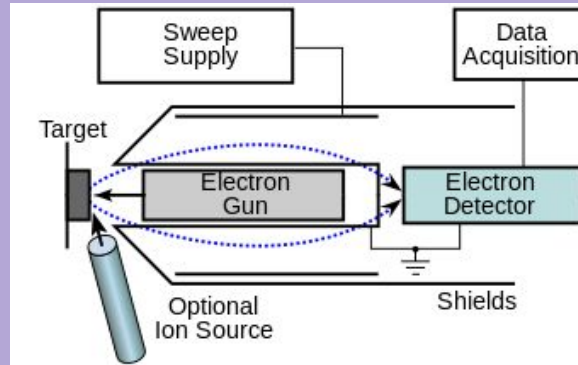
$$n \in \mathbb{N}$$

Chemical Shifts

- When chemically bonded atoms are examined, their normal peaks will be shifted.
- These shifts are experimentally determined, and can be used to understand the chemical makeup of a sample.

AES Experimental Process

- Like with XPS, the pressure must be incredibly low, at maximum $\sim 10^{-5}$ Torr.
- A beam of electrons is fired at the sample, which removes core electrons from atoms, releasing Auger electrons.
- The Auger electrons are then collected and analyzed using a CMA.
- Histograms of the electron kinetic energies are then compared with the AES spectra of materials to identify them.

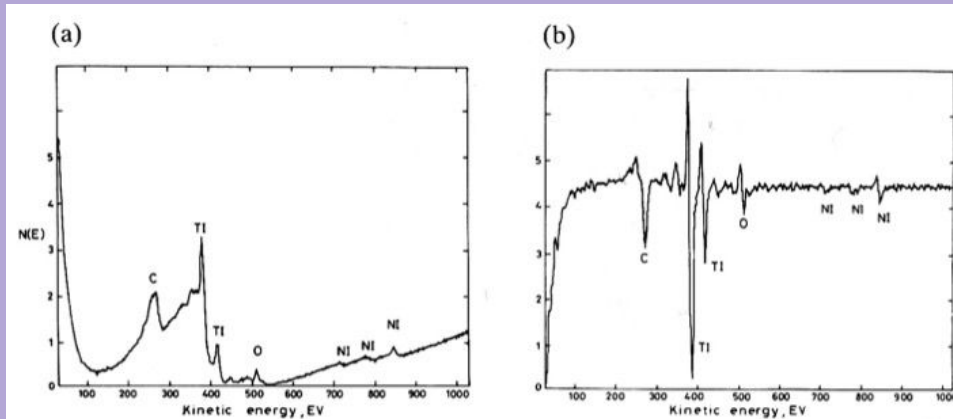


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AES Spectra Analysis

- There are three types of electrons that appear in an AES spectrum:
 - Backscattered Electrons - electrons that have been “bounced” back to the detector
 - Secondary Electrons - core electrons that have been knocked out by the electron beam
 - Auger Electrons - electrons generated by the Auger effect
- Auger electrons characterize the the material.
-

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Differentiation of collected current to give Auger peaks.
(a) is the primary signal and (b) is the differential signal.

Uses of AES

- Analyzing the composition of metals and various other materials
- Examining small sample spots, even down to the order of 10 nm
- Can achieve a higher spatial resolution than XPS
- Like XPS, it can be used for depth profiling.

Experimental Tools

- Experimental setup includes a mass spectrometer, indicating if gas is leaking into the system
- For both XPS and AES, the pressure must be kept low, so the inside of the spectrometer must be kept as a vacuum.
- Cylindrical Mirror Analyzer
 - Used to collect and analyze x-ray photoelectrons and Auger electrons
 - Consists of two cylindrical plates held at different voltages
 - Electrons will be deflected by the plates, and only a narrow amount of deflection angles are permitted to enter an electron detector.
 - Varying the voltages of the plates can change the permitted angle, allowing for a complete scan of all energies.
- Argon Ion Gun
 - Removes the top layer of a material
 - Used for cleaning
 - If a carbon peak is found, use the Ion gun to remove layers until the carbon peak disappears.

Sources

- [1]<https://mmrc.caltech.edu/XPS%20Info/Practical%20Guides%20to%20XPS/Intro%20to%20XPS.pdf>
- [2]https://mmm.edpsciences.org/articles/mmm/pdf/1995/03/mmm_1995_6_3_253_0.pdf
- [3][https://chem.libretexts.org/Bookshelves/Physical_and_Theoretical_Chemistry_Textbook_Maps/Supplemental_Modules_\(Physical_and_Theoretical_Chemistry\)/Spectroscopy/Photoelectron_Spectroscopy/Applications_of_Photoelectron_Spectroscopy](https://chem.libretexts.org/Bookshelves/Physical_and_Theoretical_Chemistry_Textbook_Maps/Supplemental_Modules_(Physical_and_Theoretical_Chemistry)/Spectroscopy/Photoelectron_Spectroscopy/Applications_of_Photoelectron_Spectroscopy)
- [4][https://chem.libretexts.org/Bookshelves/Analytical_Chemistry/Physical_Methods_in_Chemistry_and_Nano_Science_\(Barron\)/01%3A_Elemental_Analysis/1.14%3A_Auger_Electron_Spectroscopy](https://chem.libretexts.org/Bookshelves/Analytical_Chemistry/Physical_Methods_in_Chemistry_and_Nano_Science_(Barron)/01%3A_Elemental_Analysis/1.14%3A_Auger_Electron_Spectroscopy)
- [5]https://serc.carleton.edu/msu_nanotech/methods/aes.html
- [6][https://chem.libretexts.org/Courses/Franklin_and_Marshall_College/Introduction_to_Materials_Characterization_CHM_412_Collaborative_Text/Spectroscopy/X-ray_Photoelectron_Spectroscopy_\(XPS\)](https://chem.libretexts.org/Courses/Franklin_and_Marshall_College/Introduction_to_Materials_Characterization_CHM_412_Collaborative_Text/Spectroscopy/X-ray_Photoelectron_Spectroscopy_(XPS))
- [7]https://en.wikipedia.org/wiki/Photoelectric_effect
- [8]https://en.wikipedia.org/wiki/Auger_electron_spectroscopy
- [9]<https://simion.com/info/cma.html#:~:text=A%20cylindrical%20mirror%20analyzer%20>
- [10]https://catalog.libraries.wm.edu/discovery/fulldisplay?docid=alma991006335559703196&context=L&vid=01COWM_INST:01COWM_WM_NEWUI&lang=en&search_scope=WMCWF&adaptor=Local%20Search%20Engine&tab=Everything&query=any,contains,x-ray%20photoelectron%20spectroscopy&offset=0

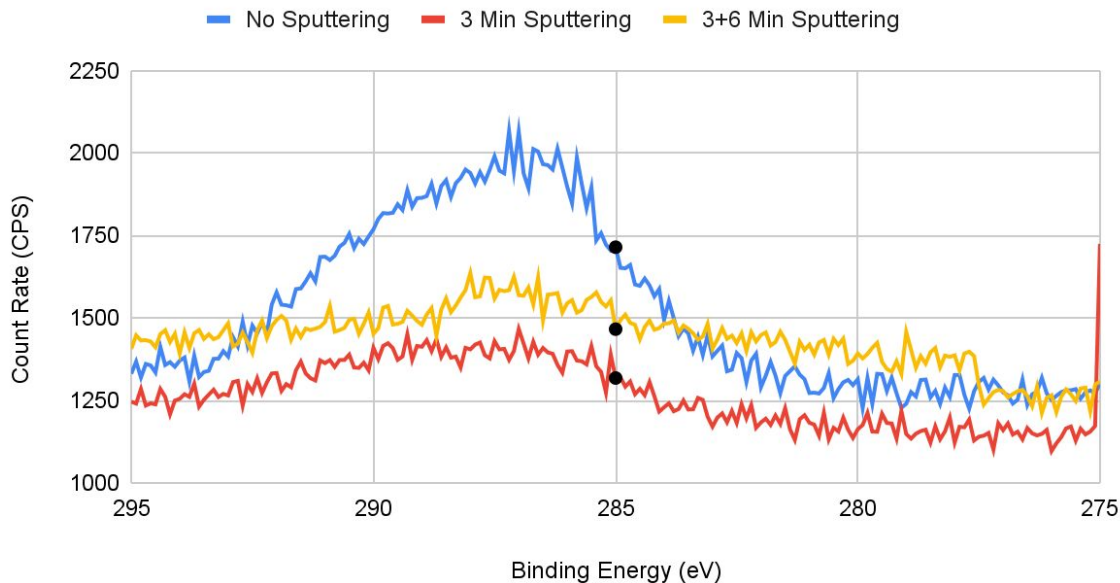
Example 1018 Carbon Steel XPS Analysis

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Initial Depth Profile of Carbon 1s Region

Scan Parameter	Value
Date	7/1/24
Peak of Interest	C1s - 285 eV
CMA Voltage	5 keV
Pass Energy	200 eV
Range	525-545 eV
Step Size	0.1 eV
Integration Time	2000 ms

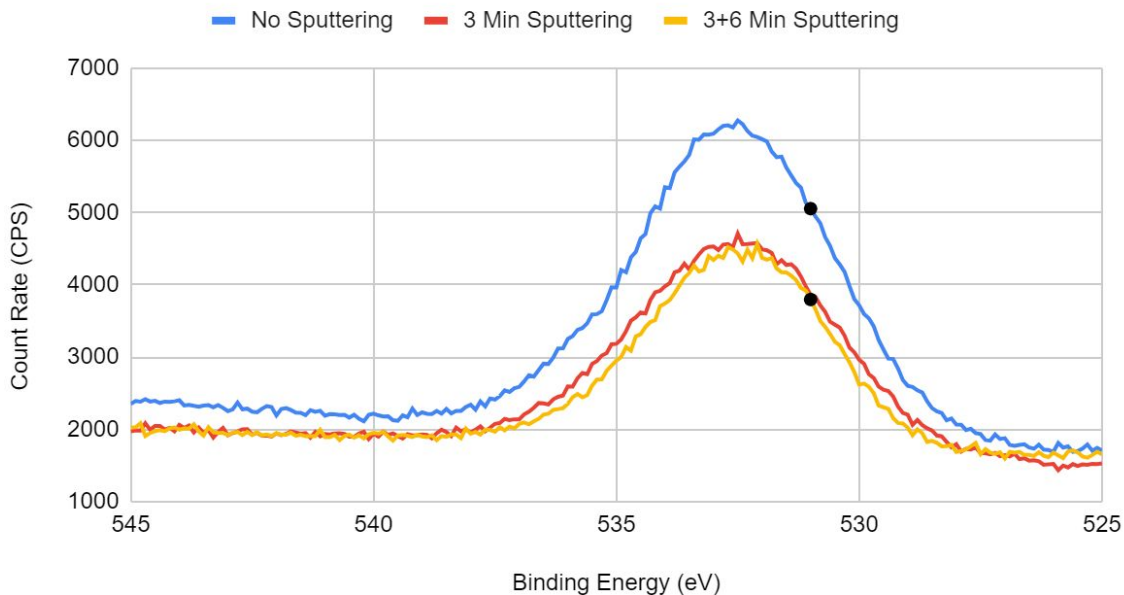
C1s Initial Depth Profile



Initial Depth Profile of Oxygen 1s Region

Scan Parameter	Value
Date	7/1/24
Peak of Interest	O1s - 531 eV
CMA Voltage	5 keV
Pass Energy	200 eV
Range	275-295 eV
Step Size	0.1 eV
Integration Time	2000 ms

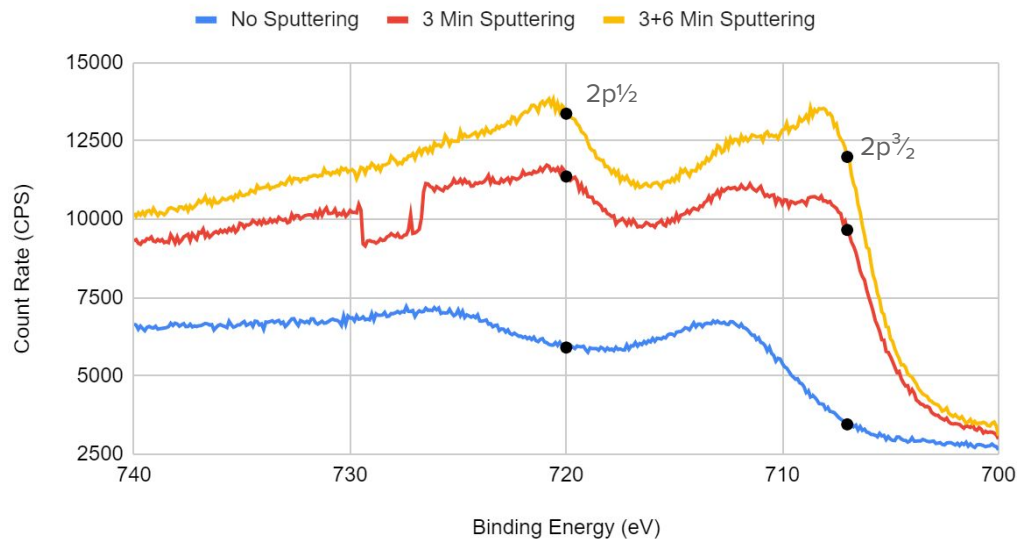
O1s Initial Depth Profile



Initial Depth Profile of Iron 2p Region

Scan Parameter	Value
Date	7/1/24
Peak of Interest	Fe2p - 707,720 eV
CMA Voltage	5 keV
Pass Energy	200 eV
Range	700-740 eV
Step Size	0.1 eV
Integration Time	2000 ms

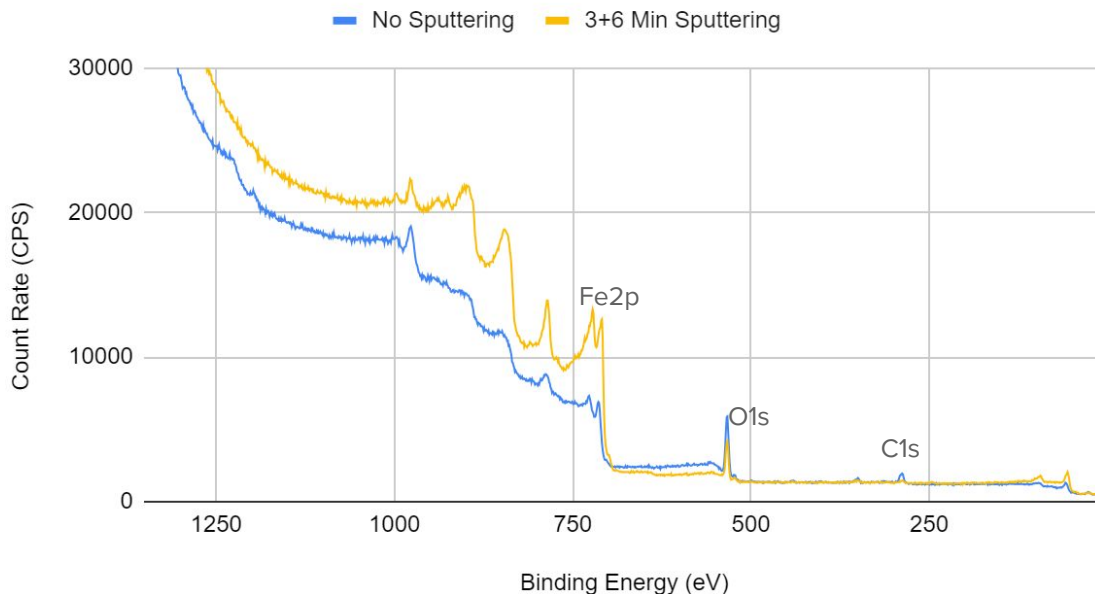
Fe2p Initial Depth Profile



Initial Depth Profile Survey Scan

Scan Parameter	Value
Date	7/1/24
Peak of Interest	All
CMA Voltage	5 keV
Pass Energy	200 eV
Range	5-1350 eV
Step Size	1 eV
Integration Time	2000 ms

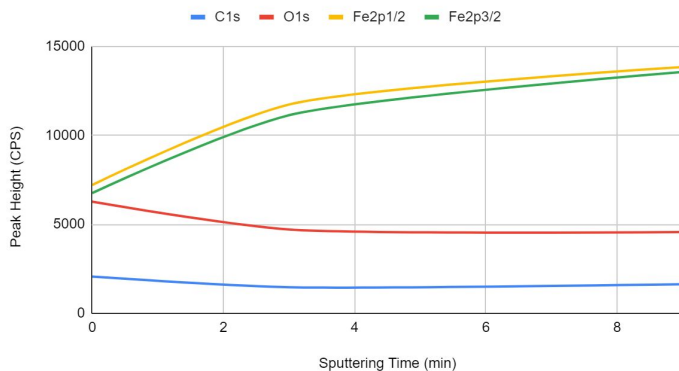
Initial Depth Profile Survey Scan



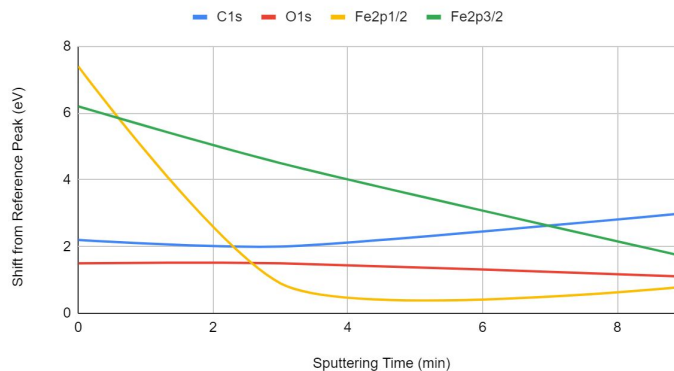
Notes on Initial Depth Profile Scans

- Using the three data points for each peak, plots for peak height and peak shift were made. These plots are likely inaccurate due to the limited amount of data, however.
- The left plot shows that the C and O peaks diminished after sputtering while both Fe peaks increase after it.
- The right plot shows that the peak shifts for both Fe peaks decreased, possibly showing the removal of the oxide layer. More data is needed.

Initial Depth Profile Peak Heights

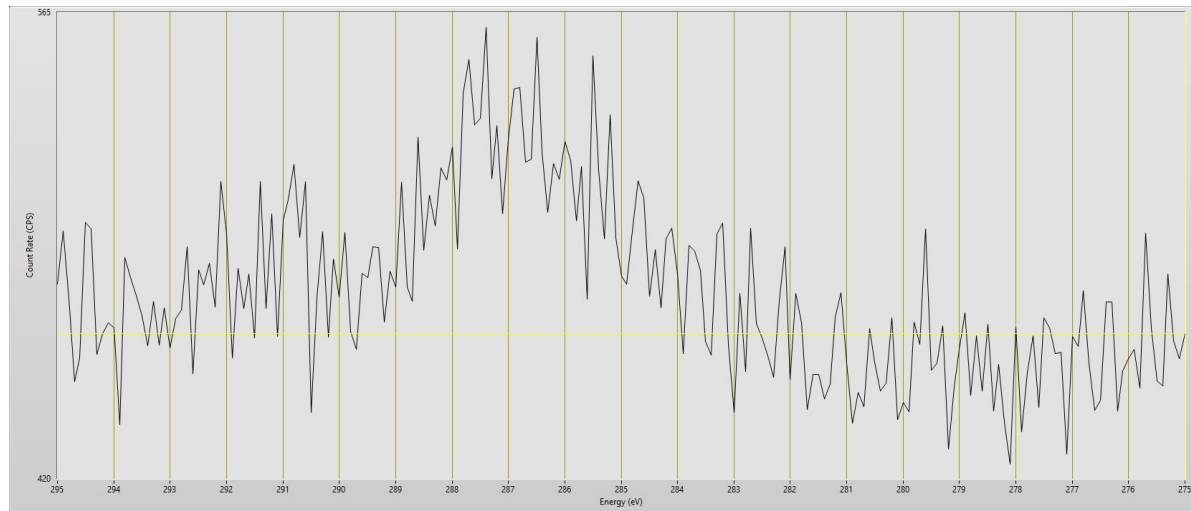


Initial Depth Profile Peak Shifts



Measuring Carbon Peak One Day After Sputtering

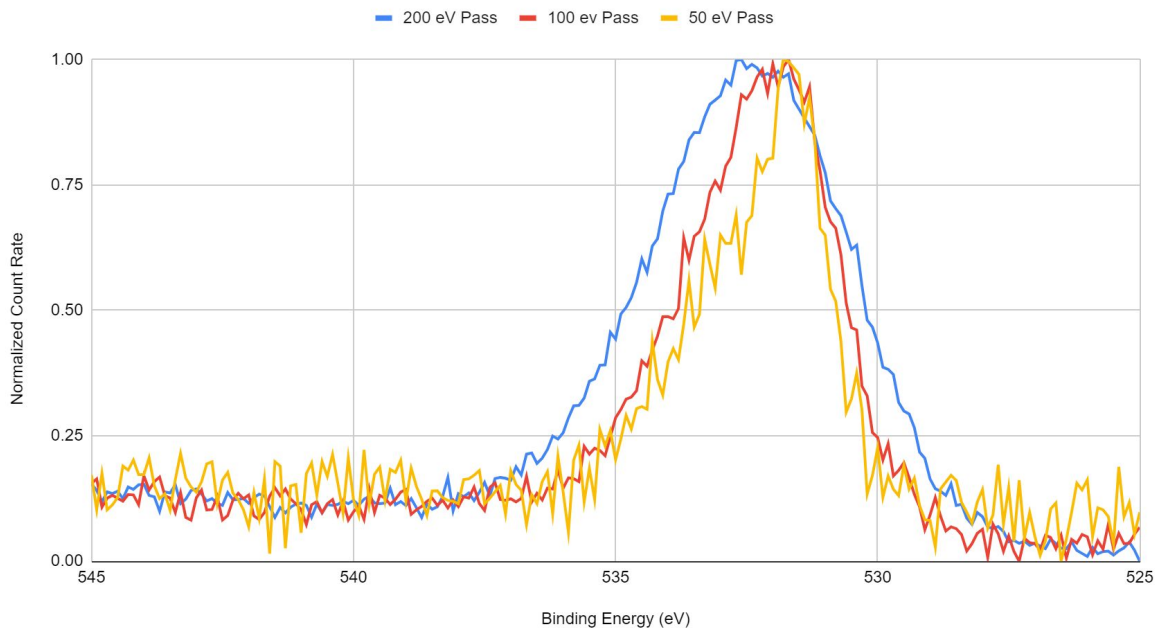
Scan Parameter	Value
Date	7/2/24
Peak of Interest	C1s - 285 eV
CMA Voltage	6 keV
Pass Energy	100 eV
Range	275-295 eV
Step Size	0.1 eV
Integration Time	2000 ms



Testing Different Pass Energies

Scan Parameter	Value
Date	7/2/24
Peak of Interest	O1s - 531 eV
CMA Voltage	6 keV
Pass Energy	200, 100, 50 eV
Range	525-545 eV
Step Size	0.1 eV
Integration Time	2000 ms

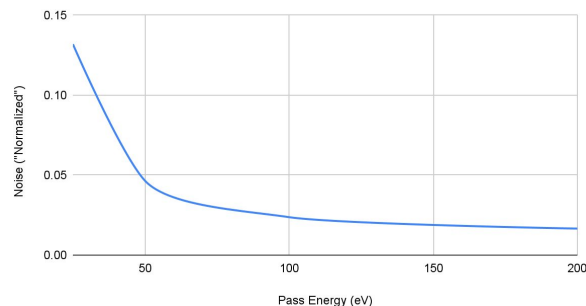
Effect of Pass Energy on 1s Oxygen Peak



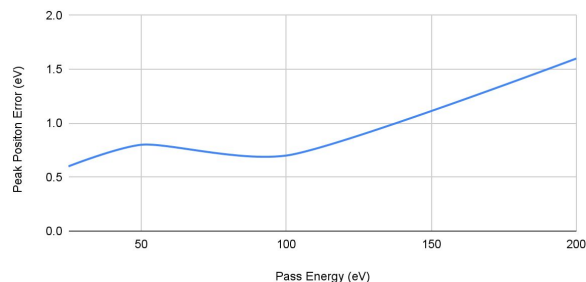
Notes on Pass Energy

- Normalizing function: $(x - \min)/(\max - \min)$
- Increasing the pass energy seems to increase the accuracy of the peak position and width, but also increases the noise.
- 25 eV pass energy was also tested, but the noise was too high to reasonably plot.
- The binding energies greater than 540 eV were considered “flat” for calculating noise.

Signal Noise vs Pass Energy in O1s Region



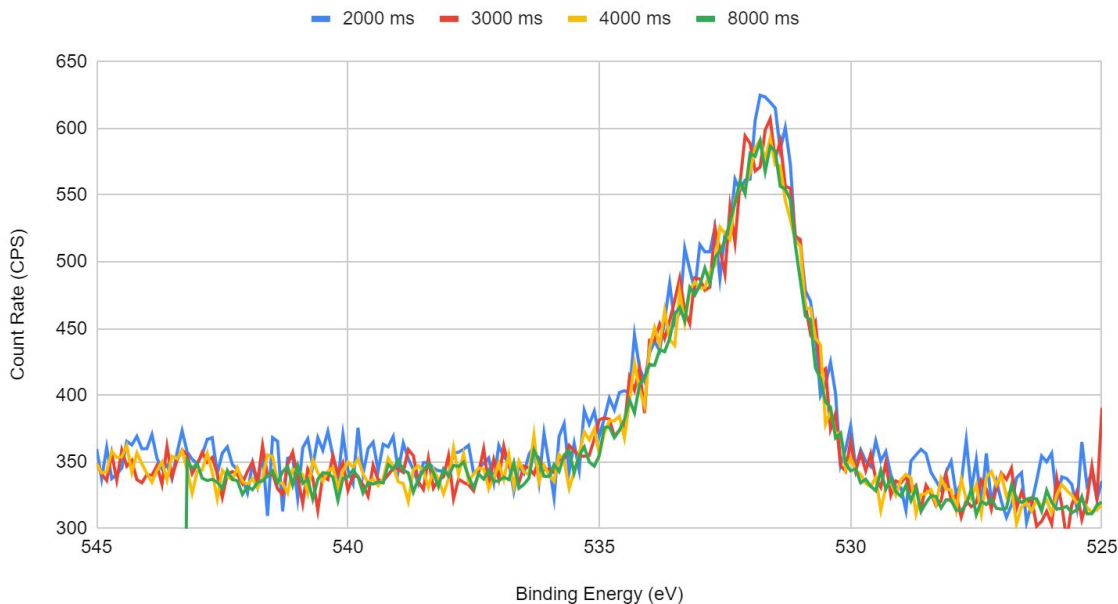
Peak Position Relative to Reference vs Pass Energy in O1s Region



Testing Different Integration Times

Scan Parameter	Value
Date	7/2/24
Peak of Interest	O1s - 531 eV
CMA Voltage	6 keV
Pass Energy	50 eV
Range	525-545 eV
Step Size	0.1 eV
Integration Time	2000, 3000, 4000, 8000 ms

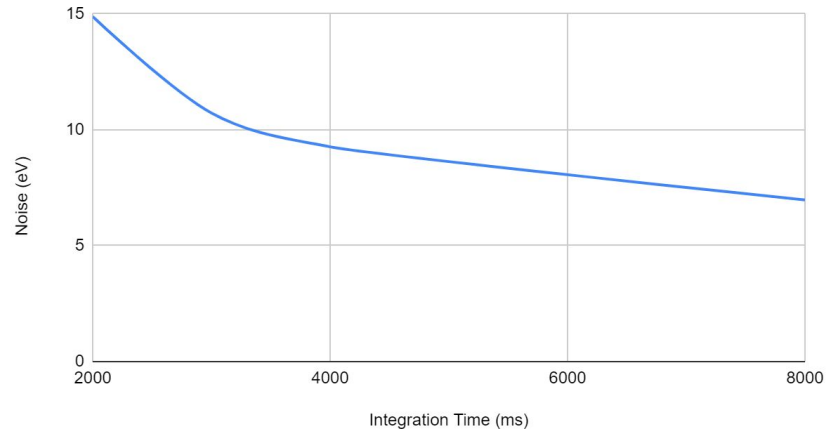
Effect of Integration Time on 1s Oxygen Peak



Notes on Integration Time

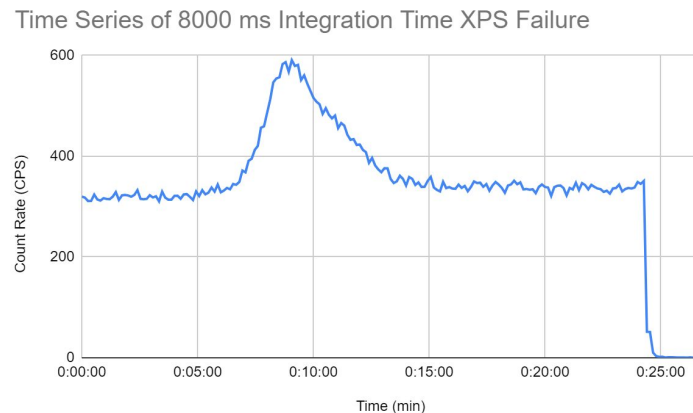
- Increasing the integration time had a clear effect of reducing the noise of the signal, however this decrease became less and less the further the integration time was increased.
- It is important to note that in the 8000 ms scan, the count rate fell to zero at around 543 eV.

Signal Noise vs Integration Time in O1s Region



Notes on the XPS Failure

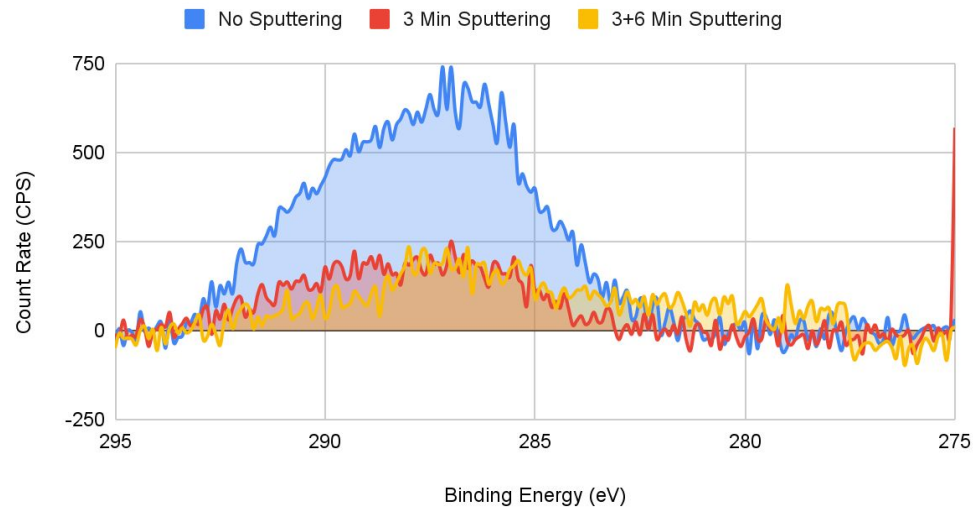
- During the final integration time test at 8000 ms on 7/2/24 around 3:00 PM, the count rate abruptly dropped to near-zero at around 543.3 eV and did not return to normal.
- As of 7/10/24, the XPS system is still broken.
- The chart below shows the drop-off during the scan.



Removing Background from Carbon 1s

Scan	Background Fit
No Sputtering	$4.69 \cdot BE^{-20.6}$
3 Min Sputtering	$5.06 \cdot BE^{-236}$
3 + 6 Min Sputtering	$7.79 \cdot BE^{-842}$

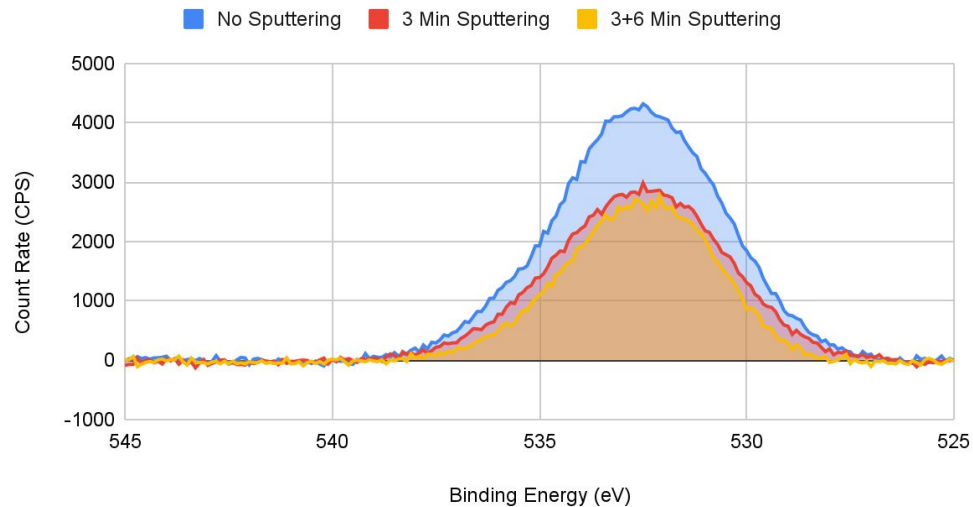
C1s Depth Profile w/out Background



Removing Background from Oxygen 1s

Scan	Background Fit
No Sputtering	32.91*BE-15567
3 Min Sputtering	26.8*BE-12545
3 + 6 Min Sputtering	17.7*BE-7622

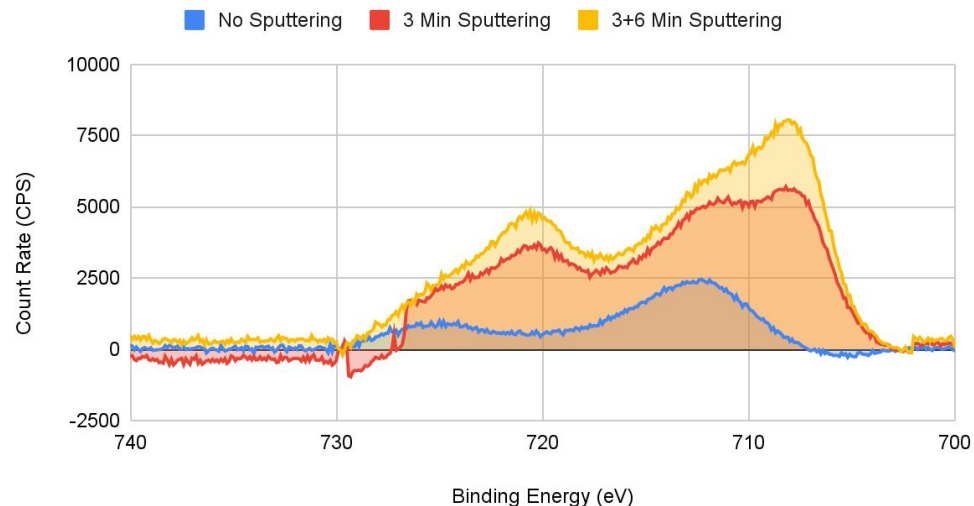
O1s Depth Profile w/out Background



Removing Background from Iron 2p

Scan	Background Fit
No Sputtering	$32.4 \cdot \text{BE} - 19957$, $144.6 \cdot \text{BE} - 98722.6$, $-23.6 \cdot \text{BE} + 23970$
3 Min Sputtering	$172 \cdot \text{BE} - 117466$, $240.7 \cdot \text{BE} - 165401.3$, $-111 \cdot \text{A}330 + 91705$
3 + 6 Min Sputtering	$137 \cdot \text{BE} - 92856$, $278.4 \cdot \text{BE} - 191645.8$, $-158 \cdot \text{BE} + 126642$

Fe2p Depth Profile w/out Background



Important Equations

- I_p is the intensity of a peak, calculated by integrating the background-subtracted count rate over the region.
$$I_p = \int_a^b (c(E) - B_p(E))dE$$
- I_X is the normalized atomic intensity, calculating the sum of the intensities of the examined peaks, and then divided by the sum of the sensitivity factors of the examined peaks.
$$\tilde{I}_X = \frac{\sum_p I_p}{\sum_p S_p}$$
- A_X is the atomic fraction, calculated by dividing the normalized atomic intensity by the sum of all the normalized atomic intensities.

$$A_X = \frac{\tilde{I}_X}{\sum_Y \tilde{I}_Y}$$

No Sputtering Scans Quantitative Analysis

	Carbon 1s	Oxygen 1s	Iron 2p1/2	Iron 2p3/2
Peak Sensitivity	1	2.93	5.6	10.82
Peak Intensity (cpsJ)	4103.898	21476.468	22810.058	
Normalized Peak Intensity (cpsJ)	4103.898	7329.853	1389.163	
Atomic Fraction	32.00%	57.16%	10.83%	

3 Min Sputtering Scans Quantitative Analysis

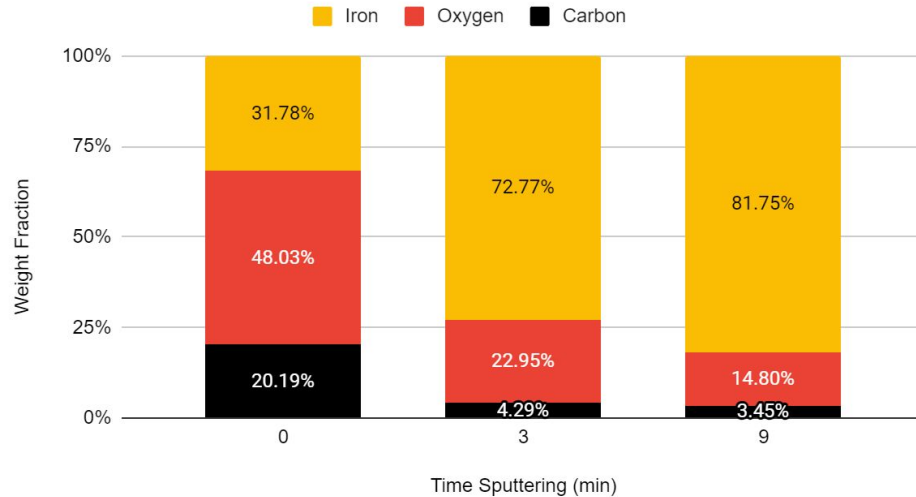
	Carbon 1s	Oxygen 1s	Iron 2p1/2	Iron 2p3/2
Peak Sensitivity	1	2.93	5.6	10.82
Peak Intensity (cpsJ)	1239.629	14590.78	74286.943	
Normalized Peak Intensity (cpsJ)	1239.629	4979.788	4524.174	
Atomic Fraction	11.54%	46.35%	42.11%	

3+6 Min Sputtering Scans Quantitative Analysis

	Carbon 1s	Oxygen 1s	Iron 2p1/2	Iron 2p3/2
Peak Sensitivity	1	2.93	5.6	10.82
Peak Intensity (cpsJ)	1256.566	11861.129	105174.935	
Normalized Peak Intensity (cpsJ)	1256.566	4048.167	6405.294	
Atomic Fraction	10.73%	34.57%	54.70%	

Quantitative Analysis Plots

Weight Fractions of 1018 Steel Depth Profile



Atomic Fractions of 1018 Steel Depth Profile

